

Week 14

Ethics and Data

Ethical Issues in Data Collection and Management

Ethics and Data Analytics

Ethical issues in Data Presentation and Dissemination

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Progress

- **Quiz 3 (Socrative)**
- **Assignment**
 - Due to Sunday
- **Project: 2nd**
 - Due to Sunday
- **Tech Ignite 2025**
 - Apr 29, 2025
 - Posters
- **Final Exams**
 - 11 May, 2025

Quiz 3 (Via Socrative)

- Room Number: **CIS6231GROUP1**



Ethics (Recap)

- Morality and Ethics are used to distinguished between good and bad
- **Morality** related to personal and normative
- **Ethics** are Rules of conduct in a particular culture or group recognized by an external source or social system.
- **Ethical Theories** is the reference
- Ethics depends on place or domain (society and culture):
 - Theoretical level ('what is the best **theory** of the good life?')
 - Practical, applied level as will be our focus ('how **should we act in this** or that situation, based upon our best theories of ethics?').

Ethics and Technology

- IEEE entire division devoted to Technology Ethics
 - IEEE conference annually
- IEEE is working on new **ethical standards** in emerging areas such as AI, robotics, and data management
- Technology increasingly shapes how human beings seek the good life
 - Well-designed/well-used technologies vs Poorly designed or misused technologies
- Many lawmakers lack the technical expertise needed to guide effective technology policy

What does ethics have to do with data?

- Ethical issues are everywhere in the world of data
- Data's collection, analysis, transmission and use impact the ability of individuals and groups to live well.
- **Big data** not just the existence and explosive growth of large digital datasets.
- New techniques, organizations, and processes involved to transform large datasets into valuable human knowledge.
- Utilize:
 - Data generation, mining, scraping, and sampling
 - Artificial intelligence and machine learning; natural language and image processing; computer modeling and simulation
 - Cloud computing and storage
 - Cybersecurity

What does ethics have to do with data?

- **Perfect storm of ethical risks**
 - increasingly powerful, misleading/misused data analytics, data-saturated
 - poorly regulated commercial environment
 - absence of widespread, well-designed standards for data practice in industry, university, non-profit, and government sectors
- Developing a **broader and better understanding** of data ethics among those who design and implement data tools and practices, is increasingly recognized as essential to meeting this goal.

What does ethics have to do with data?

- Module developed by Markkula Center for Applied Ethics at Santa Clara University
 - Part 1: The potential ethical harms and benefits presented by data
 - Part 2: Common ethical challenges faced by data professionals and users
 - Part 3: The nature and source of data professionals' ethical obligations to the public
 - Part 4: General frameworks for ethical thinking and reasoning
 - Part 5: Ethical 'best practices' for data practitioners
- No single set of ethical rules or guidelines will fit all data circumstances
- Must be adapted to the needs of many kinds of data practitioners operating of different **contexts**
- Many companies, universities, non-profit agencies, and professional societies are funding an increasing number of data ethics-related programs and training tools.
 - *Appendix A*

Discussion:

Which of these life-impacting events, both positive and negative, might be the direct result of data practices?

- **Rosalina:** a promising and hard-working law intern with a mountain of student debt and a young child to feed, is denied a promotion at work that would have given her a livable salary and a stable career path, even though her work record made her the objectively best candidate for the promotion.
- **John:** a middle-aged father of four, is diagnosed with an inoperable, aggressive, and advanced brain tumor. Though a few decades ago his tumor would probably have been judged untreatable and he would have been sent home to die, today he receives a customized treatment that in people with his very rare tumor gene variant, has a 75% chance of leading to full remission.
- **The Patels:** a family of five living in an urban floodplain in India, receive several days advance warning of an imminent, epic storm that is almost certain to bring life-threatening floodwaters to their neighborhood. They and their neighbors now have sufficient time to gather their belongings and safely evacuate to higher ground.

significant harms and benefits can data present?

- The extent to which the information is known outside the business
- The extent of measures taken by individuals possessing the trade secret to guard the secrecy of the information
- The value of the information to the owner and to the competitor
- The amount of effort or money spent by the owner to develop or gather the information.
- The ease or difficulty with which the information could be properly acquired or duplicated by others

significant harms and benefits linked to data

- Life interests
- Data has a broader ethical sweep (choices may cause harm)
- Data practitioners must confront a far more complex ethical landscape
 - Complex and different perspective

Ethically Significant Benefits Of Data Practices

- Human Understanding
 - Data can greatly enrich our understanding of ethically significant relationships
 - Data can help us to better understand how complex systems interact at a variety of scales
 - the more we understand about the world and how it works, the more intelligently we can act in it.
- Social, Institutional, And Economic Efficiency
 - Once we have a more accurate picture of how the world works, we can design or intervene in its systems to improve their functioning.
 - Example: big data help for regional traffic
- Predictive Accuracy And Personalization
 - help to make social systems work more efficiently

Ethically Significant Harms Of Data Practices

- Harms To Privacy & Security: Social, Institutional, And Economic Efficiency
- Harms To Fairness And Justice:
- Harms To Transparency And Autonomy
 - Factor risks
 - Loan approval by Deep learning? Can't explain to customer

Ethical Challenges In Appropriate Data Collection And Use

- Presented by couple of questions
- How acknowledge and respect the purpose for certain data shared/generated for us?
 - personal data shared it with me for the explicit purpose of aiding my medical research program
 - may I then sell that data to a data broker who may sell it for any number of non-medical purposes?
- How can we avoid unwarranted or indiscriminate data collection
 - Collecting more data than is justified in a particular context?
 - When is it ethical to scrape websites for public data,? Purpose?
- Ethically Considered selling or sharing subjects' data with third-parties?

Ethical Challenges In Appropriate Data Collection And Use

- Have we given data subjects appropriate forms of choice in data sharing?
 - opt-in (agreement) or opt-out (systems development, user interface) privacy settings?
 - Are data subjects 'boxed in' by the circumstances?
- Are the terms of our data policy laid out in a clear, direct, and understandable way, and accessible to all data subjects?
 - Avoid unnecessarily legalistic or technical jargon,
 - Does the design of our interface encourage careful reading ?(UX/UI)
- Are data subjects given clear paths to obtaining more information or context for a data practice?
 - buttons such 'Why am I seeing this ads?', 'How do I disable sharing'?

Ethical Challenges In Appropriate Data Collection And Use

- Are data subjects being appropriately compensated for the benefits/value of their data?
 - Value subject need?
 - Understood the interaction as we do? Exploited?
- Have we considered what control or rights our data subjects should retain over their data?
 - Should they be able to withdraw, correct, or update the data later if they choose?
- Are data subjects given clear paths to obtaining more information or context for a data practice?
 - buttons such 'Why am I seeing this ads?', 'How do I disable sharing'?

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 - Should they be able to withdraw, correct, or update the data later if they choose?
 - Who should own the shared data and hold rights over its transfer?
 - Commercial and noncommercial use?

Data Storage, Security And Responsible Data Stewardship

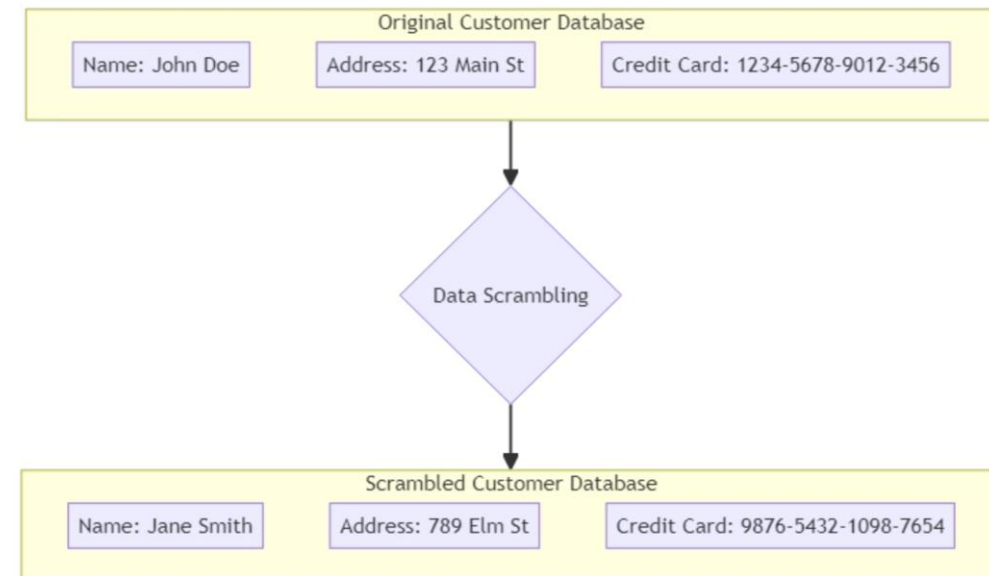
- How can we responsibly and safely store personally identifying information (PII data)? Value subject need?
 - Are subjects given clear and accurate information about our terms of storage?
 - members of our organization are responsible for which aspects of our data?
- What are concrete action plans for the worst-case-scenarios data breach?
 - mitigation strategies to limit or remedy harms (data steward for plan)
- An appropriate investments in our data security/storage infrastructure?
 - Harms occurred by allocating insufficient
 - Contracting with unreliable/low-quality data storage and security vendors?
- What privacy-preserving techniques we rely on?
 - Data anonymization
 - Obfuscation (In the event of a data breach, sensitive data will be useless to attackers)
 - Scrambling data

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 - <https://mostly.ai/blog/data-anonymization-in-python>
 - Scrambling data

Data Storage, Security And Responsible Data Stewardship

- Do we have an end-to-end plan for the lifecycle of the data we collect or used
 - examine that plan to see if it needs to be improved or updated
 - Data steward (Data governance)
- Measures/standards/criteria should we have to delete, correct, or update
 - Easily accessible by affected/interested parties



By Data Sunrise :

<https://www.datasunrise.com/knowledge-center/data-scrambling/>

Data Hygiene And Data Relevance

- What are our practices and procedures for validation and auditing of data in our context,
 - data conform to the necessary constraints of our data practice
- How do we establish proper parsing and consistency of data field labels
 - integrating data from different sources/systems/platforms?
 - How do we ensure the integrity during transfer?
- What are our established tools and practices for scrubbing dirty data
- How long is this data likely to remain accurate, useful or relevant?
 - What is our plan for replacing/refreshing datasets that have become out-of-date?

Extra Reading: Deep Sync Website: What Is Data Hygiene? + 6 Best Practices for Accurate Data

- <https://deepsync.com/data-hygiene/>

Validation And Testing Of Data Models & Analytics

- How to ensure that we have adequately tested our analytics/data models to validate their performance
- How to consider the ethical harms that may be caused by validation and testing
- Ethical challenges might arise from the lack of transparency
 - Deep learning (Black box)
- How to test data analytics and models to ensure their reliability across new, unexpected contexts?
 - drug has approved to treat a condition different than your condition
- How ensure that the results, applications, or other consequences of our analytics are audited for disparate and unjust outcomes?
 - How respond if systems or data practices are accused by other outcomes, or other social harms?

Human Accountability In Data Practices And Systems

- Who will be designated as responsible for each aspect of ethical data practice?
 - specific individuals are made responsible for taking action
- Who should and will be held accountable for harms
 - “problem of many hands ”
 - AI or model bias not for programmer, but from data from different resources and places
- Did we establish effective organizational or team practices and policies
- Do we have a clear and effective process for any harmful outcomes of our data practice
- To what extent should our data systems and practices be open for public inspection and comment?

Understanding Personal, Social, And Business Impacts Of Data practice

- data/ system impact data subjects or other parties
 - sufficiently diverse to understand and anticipate its effects of data/systems
- sufficient input gathered from other stakeholders
 - who might represent very different interests/values/experiences from ours?
- Data collection violating legal or moral rights
 - Limit human capabilities
 - impinge on the autonomy or dignity of other moral agents?
 - damage or interfere with the moral and intellectual habits, values
- Data practice may be controversial or damaging to professional reputation of those involved.

Data Practitioners' Obligations To The Pub

- Considering how our data/ system impact data subjects or other parties later
 - sufficiently diverse to understand and anticipate its effects of data/systems
- sufficient input gathered from other stakeholders
 - who might represent very different interests/values/experiences from ours?
- Has the impact of testing of the practice taken into account
 - across a variety of individuals, identities, cultures and interest groups?
- Does data collection violate legal or moral rights
 - Limit human capabilities
 - impinge on the autonomy or dignity of other moral agents?
 - damage or interfere with the moral and intellectual habits, values
- Data practice may be morally or socially controversial or damaging to professional reputation of those involved.

Best Practices For Data Ethics

- Best Practices For Data Ethics
 - Keep Data Ethics in the Spotlight—and Out of the Compliance Box
 - Consider the Human Lives and Interests Behind data.
 - Focus on Downstream Risks and Uses of Data.
 - Mind the Gap Between Expectations and Reality.
 - Treat Data as a Conditional Good.
 - Establish Chains of Ethical Responsibility and Accountability.
 - Design for Privacy and Security
- Best Practices For Living Well
 - Look for Moral Exemplars:
 - Exercise Moral Imagination:
 - Practice Self- Reflection/Examination
 - Acknowledge Our Own Moral Strength
 - Seek the Company of Other Moral Persons